



Addis Ababa Science and Technology University

Department of Electromechanical Engineering

Automated Bag Sealing System (ABSS)

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Abstract

This report presents the design and proposal of an Automated Bagging and Sealing System (ABSS) tailored for Bejai Ethio Industrial and Engineering Solutions. The system integrates a 50 kg load cell, auger filler, conveyor belt, and continuous band sealer under PLC-based control. Mechanical, electrical, and software components were modeled and analyzed to ensure accurate weighing, efficient material transfer, and consistent sealing of 25 kg woven polypropylene (WPP) bags laminated with biaxially oriented polypropylene (BOPP) film. Calculations were carried out for conveyor speed, auger torque, and sealing dwell time to determine appropriate motor ratings and operational parameters. The proposed system demonstrated significant improvements compared to the existing manual process, including reduced labor requirements, enhanced accuracy, higher throughput, and better packaging quality. While implementation was limited to conceptual design during the internship, the study highlights a feasible and impactful engineering solution to an identified industrial problem.

Keywords: Automated bagging, Heat sealing, Conveyor system, Load cell, PLC control, Packaging automation

Contents

1	Introduction to Bejai Ethio Industrial & Engineering Solution	1
1.1	History of Hosting Company	1
1.2	Services Offered	1
1.3	Products	1
1.4	Key Customers of the Company	2
1.5	Company Organization and Workflow	2
2	Overall Internship Experiences	3
2.1	Working Experience	3
2.2	Workflow of the Company	3
2.3	Working Sector	4
2.4	Challenges Faced During the Internship Program	4
2.5	Solutions for the Problems	4
2.6	Conclusion	4
3	Selected Industrial Project	5
3.1	Design Project Title	5
3.1.1	Short Summary	5
3.2	Problem Statement and Justification	5
3.3	Objective of the Project	5
3.4	Methodology	6
3.4.1	Observation and Problem Identification	6
3.4.2	System Design Approach	6
3.4.3	Mechanical Design	7
3.4.4	Electrical Design	10
3.4.5	Software Design	12
3.4.6	Literature Review	13
3.5	Results and Discussion	14
3.6	Results	14
3.7	Discussion	15
3.8	Conclusion	15
4	Overall Internship Achievements and Experiences	16
4.1	Theoretical Knowledge Enhancement	16
4.2	Practical Skill Development	16
4.3	Industrial Problem-Solving Capability	17
4.4	Teamwork and Collaboration Skills	17
4.5	Understanding of Work Ethics	18
4.6	Entrepreneurship and Industry Insight	18
5	Conclusion	18
5.1	Recommendations	19

List of Figures

3.1	Conveyor design	8
3.2	Auger filler	9
3.3	Band sealer	9
3.4	ladder diagram	13
3.5	Final Assembly Design	15

1 Introduction to Bejai Ethio Industrial & Engineering Solution

1.1 History of Hosting Company

Bejai Ethio Industrial & Engineering Solution was established with the dream of satisfying the growing demand for industrial engineering, manufacturing, and technology solutions in Ethiopia. The company has dedicated itself to bridging the gap between international innovations and domestic demand by offering quality engineering products and services that satisfy the Ethiopian and regional markets.

Founded by engineers and businesspeople with profound technical backgrounds, Bejai Ethio began life as a small engineering consultancy. The company gradually diversified its activities over time to include industrial machinery supply, systems integration, and turnkey project execution. This growth was driven by an enthusiasm for innovation, reliability, and customer-centric solutions.

Today, Bejai Ethio is one of Ethiopia's emerging movers in the industrial and engineering sector. Its portfolio includes supporting manufacturing factories, infrastructure development, and industrial modernization initiatives. The company proudly claims to have entwined global standards with local skills, thereby contributing to the industrialization and economic growth of Ethiopia.

1.2 Services Offered

Bejai Ethio Industrial & Engineering Solution offers a wide range of services with the objective of helping industries achieve efficiency, reliability, and sustainability. These include:

- **Engineering Consultancy and Design** – Feasibility studies, system design, and optimization in mechanical, electrical, and civil engineering.
- **Industrial Maintenance and Repair** – Preventive and corrective maintenance for equipment, machinery, and production lines to ensure continuous operations and reduce downtime.
- **Project Management** – End-to-end management, including planning, execution, monitoring, and evaluation.
- **System Integration** – Integrating mechanical, electrical, and digital systems for maximum industrial performance.
- **Technical Training and Capacity Building** – Training workshops and programs to equip client workforces with modern technical skills.
- **Energy Efficiency and Sustainability Consultancy** – Energy audits, renewable energy integration, and waste reduction strategies.

1.3 Products

In addition to services, Bejai Ethio offers a wide range of industrial products and equipment from well-known manufacturers. These include:

Industrial Machinery

CNC machines, lathes, milling machines, pumps, compressors, hydraulic systems, and processing machinery.

Electrical and Electronic Equipment

Control panels, PLC systems, automation devices, power generators, distribution systems, and industrial lighting/safety equipment.

Engineering Supplies and Materials

Mechanical spares, welding and fabrication machines, bearings, seals, and industrial lubricants.

Custom Fabricated Parts

Designed and manufactured locally to meet individual client requirements, enabling cost-effectiveness and flexibility.

1.4 Key Customers of the Company

Bejai Ethio serves a diverse set of customers, including:

- **Manufacturing Industries** – Textile, food and beverage, chemical, cement, and metal plants.
- **Construction Companies** – Infrastructure development projects requiring engineering solutions and equipment.
- **Energy and Utility Providers** – Power generation, transmission, and renewable energy projects.
- **Governmental and Public Institutions** – Industrial development projects in partnership with public entities.
- **Small and Medium Enterprises (SMEs)** – Low-cost solutions for startups and growing businesses.

1.5 Company Organization and Workflow**Organizational Structure**

- **Board of Directors / Founders** – Provide strategic leadership and company vision.
- **General Manager** – Directs operations, decision-making, and coordination.
- **Engineering & Technical Department** – Consultancy, design, and technical services.
- **Maintenance & Operations Department** – Service delivery, repair, and maintenance.

- **Procurement & Supply Chain Department** – Manages sourcing, supply, and logistics.
- **Sales & Marketing Department** – Customer relations and business development.
- **Administration & Finance Department** – Human resources, finance, and support services.

Workflow System

1. Customer Engagement – Initial consultations to understand needs.
2. Requirements Assessment & Proposal – Business and engineering analysis of solutions.
3. Project Execution / Service Delivery – Implementation by expert teams.
4. Quality Assurance & Monitoring – Continuous checks to ensure safety and compliance.
5. After-Sales Support – Training, technical assistance, and maintenance.

2 Overall Internship Experiences

2.1 Working Experience

The internship at Bejai Ethio Industrial & Engineering Solutions proved to be a highly rewarding and productive experience. The workplace environment was welcoming, and it gave us the opportunity to observe how professional engineers and technicians carry out their daily responsibilities. A large portion of our time was spent in the office, where we worked on 3D part modeling and technical drawings using SolidWorks and AutoCAD. This provided us with valuable hands-on experience in digital design and exposed us to the kind of precision and attention to detail required in real engineering practice.

Beyond the office setting, we also visited the company's workshop. There, we toured different machines and production systems. While the office work allowed us to conceptualize and design machine parts virtually, the workshop visits helped us witness how these components are actually manufactured, assembled, and integrated into real operations. This direct connection between design and fabrication strengthened our understanding of how theoretical concepts translate into practical, functional equipment.

2.2 Workflow of the Company

From our observation, the company follows an integrated workflow that combines office-based design with practical workshop execution. Engineers begin by developing ideas and creating precise 3D models and technical drawings based on customer requirements. Once finalized, these designs are transferred to the workshop, where the actual machines or parts are fabricated using the available equipment. In addition to manufacturing, the company also provides installation and maintenance services, ensuring that their solutions remain fully operational at customer sites.

Although our main focus was on the design side, the workshop tours made it clear how essential both stages are. Seeing how a design created on a computer screen transforms into a working machine gave us a holistic view of engineering practice, reinforcing the importance of collaboration between design and manufacturing teams.

2.3 Working Sector

Bejai Ethio Industrial & Engineering Solutions is engaged in a broad range of engineering fields. We observed that the company designs and manufactures machines for animal feed plants, wheat flour processing, and other specialized industrial systems. Its well-equipped workshop allows it to produce a diverse array of machinery and parts, making it one of the key engineering solution providers in the local industry.

Observing such a variety of machines and systems under one roof was enlightening. It gave us a sense of how versatile and diverse the engineering profession can be, and how a single company can provide solutions across multiple industries.

2.4 Challenges Faced During the Internship Program

In general, the internship environment was supportive, with cooperative supervisors and a conducive work setting. While we did not face major personal challenges, we did identify operational inefficiencies. One notable example was in the wheat flour processing line, where bag filling and sealing were performed manually. This required significant physical effort from workers and consumed considerable time, making it a bottleneck in the production process.

2.5 Solutions for the Problems

To address the issue of manual bag sealing, we proposed an automated solution. Our idea was to integrate a conveyor system with a continuous bag sealer. Under this arrangement, once a bag is filled from the silo, it would be automatically transferred by the conveyor to the sealing unit, where the bag would be heat-sealed without manual intervention.

This solution would reduce labor intensity, improve the speed of packaging, and ensure consistent sealing quality. It also demonstrated the broader role of automation in modern industry, where repetitive manual tasks can be streamlined to save time, effort, and costs.

2.6 Conclusion

In conclusion, the internship at Bejai Ethio Industrial & Engineering Solutions offered a comprehensive learning experience. It allowed us to strengthen our design skills, particularly in 3D modeling, while also exposing us to real-world workshop operations. The balance between office work and hands-on observation gave us both technical knowledge and practical insight into industrial processes.

Most importantly, we learned to identify real operational challenges, such as the inefficiencies of manual sealing, and propose engineering solutions to address them. By doing so, we not only applied our academic knowledge but also contributed ideas that could enhance the company's productivity. The skills, experiences, and problem-solving mindset developed during this internship will undoubtedly play a vital role in shaping our careers as future engineers.

3 Selected Industrial Project

3.1 Design Project Title

Design and Proposal of an Automated Bagging and Sealing System for Animal Feed Packaging

3.1.1 Short Summary

This project focuses on designing an Automated Bagging and Sealing System (ABSS) tailored for Bejai Ethio Industrial & Engineering Solutions. The company currently packages animal feed products manually, requiring several workers to fill, weigh, and stitch polypropylene bags. The manual process is slow, labor-intensive, inconsistent in weight accuracy, and prone to poor sealing quality, which leads to spillage, dust leakage, and reduced product appeal.

The proposed system integrates a load cell-based weighing unit, an auger filler for precise dispensing, a conveyor for bag transfer, and a continuous band sealer for durable closure. By automating the final stage of the packaging process, the system reduces labor requirements, improves accuracy, ensures hygienic sealing, and increases throughput.

3.2 Problem Statement and Justification

At Bejai Ethio Industrial & Engineering Solutions, most stages of the animal feed production process are semi-automated, yet the packaging stage remains largely manual. Workers are responsible for weighing the product, filling the bags, and sealing them using portable sewing machines. This approach presents several challenges:

- **Low Efficiency:** Manual packaging achieves only about one bag per minute, creating a bottleneck compared to faster upstream processes.
- **Inaccuracy in Weighing:** Manual filling leads to inconsistent bag weights, causing product loss and reducing customer confidence.
- **Poor Seal Quality:** Stitching produces variable closure quality, which often results in dust leakage, contamination, and unattractive packaging.
- **Labor Dependence:** The process requires 3–4 workers per station, increasing labor costs and making the system vulnerable to downtime when workers are fatigued or absent.

Given these limitations, automating the bagging and sealing stage is necessary to enhance efficiency, product quality, and overall productivity. The project is justified by its potential to reduce operational costs, increase throughput, and align the company with modern packaging standards.

3.3 Objective of the Project

General Objective

To design and propose an Automated Bagging and Sealing System that integrates weighing, filling, conveying, and sealing operations, thereby improving packaging efficiency and product quality at Bejai Ethio Industrial & Engineering Solutions.

Specific Objectives

1. To reduce packaging time by designing a system capable of sealing three 25 kg bags per minute.
2. To ensure accuracy in bag weight using a load cell and auger filler system.
3. To improve seal quality by replacing manual stitching with a continuous band sealer.
4. To reduce labor requirements from 3–4 workers to a single operator.
5. To minimize product loss, dust leakage, and contamination during packaging.

3.4 Methodology

The methodology adopted for this project involved observing the existing manual bagging and sealing system at Bejai Ethio Industrial & Engineering Solutions, identifying its limitations, and designing an Automated Bagging and Sealing System (ABSS). The proposed system integrates weighing, filling, and continuous band sealing to enhance productivity, ensure consistent product quality, and reduce labor dependency.

3.4.1 Observation and Problem Identification

During the internship, the packaging process was observed in detail. Bagging and sealing of animal feed products are currently performed manually. Workers fill woven polypropylene (WPP) bags laminated with BOPP film by hand, and sealing is performed using portable sewing machines. This requires three to four workers per station and results in:

- Low throughput compared to upstream processes (≈ 1 bag/min).
- Inconsistent bag weights due to manual measurement.
- Poor sealing quality, leading to spillage, contamination, and unattractive packaging.
- Heavy physical workload on workers, resulting in fatigue and downtime.
- Production stoppages whenever operators take breaks or are absent.

From these observations, the need for an **automated bagging and sealing solution** was identified.

3.4.2 System Design Approach

The design methodology followed a structured approach consisting of the following steps:

1. **Requirement Definition** – Determining system capacity (25–50 kg bags), throughput (3 bags/min), sealing method (continuous band sealer), and power source (220 V).
2. **Conceptualization** – Designing a system that integrates a hopper and auger filler with a load cell, a conveyor for bag transfer, and a continuous band sealing mechanism.

3. **Component Sizing** – Performing mechanical design and calculations for the conveyor, auger, sealer, and weighing system.
4. **Integration and Control** – Developing considerations for synchronization of weighing, filling, conveying, and sealing.
5. **Validation** – Comparing calculated capacities with required performance and including safety factors.

3.4.3 Mechanical Design

Conveyor Design The conveyor is required to transport filled bags (25–50 kg) from the filling station to the sealing unit.

- Bag mass, $m = 25 \text{ kg}$
- Gravitational acceleration, $g = 9.81 \text{ m/s}^2$
- Bag pitch, $p = 3.00 \text{ m}$
- Cycle time, $t = 20 \text{ s}$

Conveyor speed:

$$v = \frac{p}{t} = \frac{3.00}{20} = 0.150 \text{ m/s}$$

Bag weight:

$$W = m \cdot g = 25 \times 9.81 = 245.25 \text{ N}$$

Frictional force ($\mu = 0.30$):

$$F = \mu \cdot W = 0.30 \times 245.25 = 73.58 \text{ N}$$

Torque at drive pulley (radius $r = 0.05 \text{ m}$):

$$T = F \cdot r = 73.58 \times 0.05 = 3.68 \text{ N}\cdot\text{m}$$

With a safety factor of 3.0:

$$T_{\text{req}} = 3.68 \times 3.0 = 11.04 \text{ N}\cdot\text{m}$$

Steady-state power:

$$P = F \cdot v = 73.58 \times 0.150 = 11.04 \text{ W}$$

Result: Select a geared motor (220 V) with output torque $\geq 12\text{--}15 \text{ N}\cdot\text{m}$ and power rating of 0.18–0.37 kW.

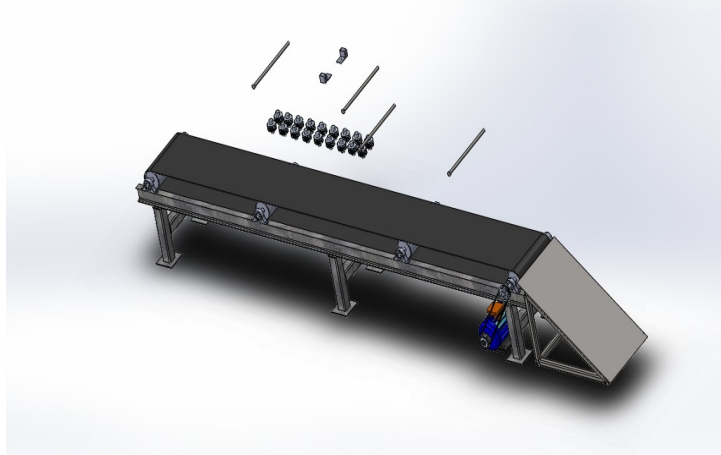


Figure 3.1: Conveyor design

Auger Filler Design The auger filler dispenses the required product weight (25 kg) into each bag.

- Bulk density of feed, $\rho = 600 \text{ kg/m}^3$

Bag volume:

$$V_b = \frac{25}{600} = 0.0417 \text{ m}^3$$

- Auger diameter, $D = 0.12 \text{ m}$
- Pitch, $p = 0.08 \text{ m}$
- Fill factor, $k_f = 0.30$

Auger cross-sectional area:

$$A = \frac{\pi D^2}{4} = 0.0113 \text{ m}^2$$

Volume per revolution:

$$V_{\text{rev}} = A \cdot p \cdot k_f = 0.000271 \text{ m}^3$$

Required revolutions per bag:

$$N_{\text{rev}} = \frac{V_b}{V_{\text{rev}}} = \frac{0.0417}{0.000271} \approx 154 \text{ rev}$$

Shaft speed for a 20 s cycle:

$$\text{RPM} = \frac{154}{20} \times 60 \approx 461 \text{ rpm}$$

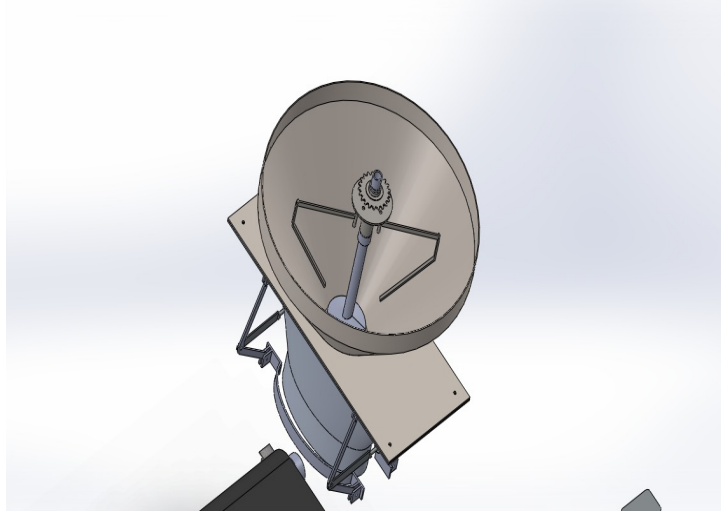
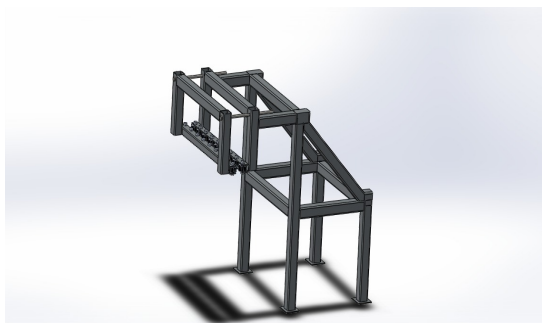


Figure 3.2: Auger filler

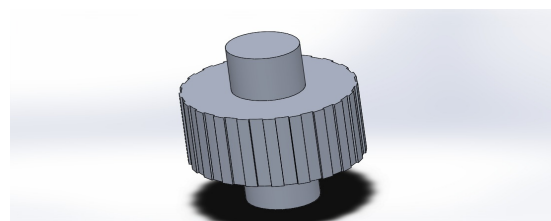
Result: The auger requires $\approx 450\text{--}500$ rpm. A $0.25\text{--}0.5$ kW geared motor with torque $\geq 15\text{--}20$ N·m is recommended, with a safety factor for product variability.

Band Sealer Design

- Type: Horizontal continuous band sealer with heated blocks and PTFE belts; independent band drive.
- Adjustables (mechanical): Temperature window suitable for BOPP-laminated WPP liners.
- Sealer surface (heater) temperature — start at: 170°C (Set PID controller to 170°C as the initial value.)
- Band speed to achieve proper thermal dwell (heated length / band speed).
- Cooling: Post-seal cooling block or air knife to fix the seal before handling.
- Dwell time (contact time) - start at: $1.2 - 1.5$ s



(a)



(b)

Figure 3.3: Band sealer

Seal quality plan: Start with vendor-recommended temp range for BOPP liners; mechanically ensure constant band tracking (band alignment screws, crowned band pulleys), then finalize pressure via test strips (peel test).

Load Cell Sensor Load cell selected: 50 kg single-point load cell.

Target accuracy: $\pm 1\%$ 25 kg $\rightarrow \pm 0.25$ kg.

To preserve accuracy mechanically:

- Isolate weighing platform from conveyor vibrations (use guide rails so bag vertical load transfers to load cell, lateral loads removed).
- Limit platform travel/deflection under full bag to ≤ 0.5 mm. That ensures the load cell output is stable and repeatable.

3.4.4 Electrical Design

The electrical design of the Automated Bagging and Sealing System (ABSS) was carried out to ensure safe, reliable, and efficient operation of all electromechanical components. The design integrates the weighing unit (load cell), auger filler motor, conveyor drive, and continuous band sealer with a centralized control panel for monitoring and operation.

Power Supply The system is powered by a single-phase 220 V AC supply, which is standard in industrial facilities in Ethiopia. The main supply passes through:

- A main circuit breaker for overload and short-circuit protection.
- Individual fuses for each subsystem (conveyor motor, auger motor, sealer heaters, and control unit).
- An emergency stop switch wired in series to cut power to the entire system during unsafe conditions.

Control Unit A Programmable Logic Controller (PLC) or microcontroller-based system is used as the central controller. The PLC receives inputs from sensors and the weighing system, processes the data, and sends commands to actuators. The functions of the control unit are:

- Control auger motor operation based on target weight signal from the load cell.
- Start and stop the conveyor motor to transport filled bags to the sealer.
- Synchronize conveyor movement with the continuous band sealer speed.
- Trigger alarms or status indicators (LED or buzzer) for underweight, overweight, or fault conditions.

Sensors and Input Devices

- **Load Cell:** A 50 kg single-point stainless-steel load cell measures the bag weight. Its analog millivolt signal is amplified and digitized using an ADC (e.g., HX711) connected to the PLC.
- **Photoelectric Sensor:** Positioned at the filling station to detect the presence of a bag before the auger starts dispensing.

- **Limit Switches:** Placed at conveyor ends for position feedback and overload protection.
- **Temperature Sensor (Thermocouple/RTD):** Embedded in the band sealer heater to regulate sealing temperature through a PID controller.

Actuators and Output Devices

- **Conveyor Motor:** A 0.18–0.37 kW geared motor (220 V) controlled via a motor starter or VFD to allow adjustable speed.
- **Auger Motor:** A 0.25–0.5 kW geared motor with torque capacity of 15–20 N·m. Its operation is started and stopped automatically based on the load cell signal.
- **Band Sealer:** Continuous band sealer with independent drive motor and heater system, temperature controlled via a solid-state relay (SSR) and PID controller.
- **Indicators:** Tower light (green, yellow, red) and buzzer for machine status, errors, or maintenance alerts.

Control Logic The sequence of operation is as follows:

1. Bag presence is detected by the photoelectric sensor.
2. The auger motor is activated, dispensing material until the load cell measures 25 kg $\pm 1\%$.
3. Once the target weight is achieved, the auger stops automatically.
4. The conveyor motor is engaged, moving the filled bag toward the band sealer.
5. The band sealer closes the bag while the temperature and dwell time are maintained within set parameters.
6. The sealed bag is discharged at the end of the conveyor, and the system resets for the next cycle.

Safety Features To ensure operator and equipment safety:

- Emergency stop switches are located at accessible points.
- Overload relays protect motors from excess current.
- Proper earthing is provided for all metallic components.
- Heat insulation and guards are installed on the sealing section to avoid accidental burns.

Result: The electrical design ensures that the ABSS operates reliably, with synchronized control of filling, conveying, and sealing, while maintaining operator safety and product quality.

3.4.5 Software Design

The software design of the Automated Bagging and Sealing System (ABSS) defines the logic by which the PLC coordinates the sequence of operations between sensors, actuators, and the user interface. The main control objective is to ensure accurate weighing, reliable filling, and consistent sealing of 25 kg bags, while minimizing operator involvement.

The PLC is programmed to continuously monitor the load cell output through an analog-to-digital converter (HX711 module). The measured weight is compared against a preset setpoint of 25 kg. The auger motor remains activated until the target weight is achieved. Once the target weight is reached, the auger stops automatically, and the conveyor motor is energized to transfer the filled bag toward the sealing unit.

The sealing unit consists of a continuous band sealer controlled by a Thermostat temperature controller. The software logic ensures that the heater achieves and maintains the set temperature (170 °C) before enabling sealing. If the temperature falls outside the allowable tolerance, the sealer is disabled and an alarm is triggered.

The sequence of operations can be summarized as follows:

1. Initialization: System power on, PLC self-check, sensors reset.
2. Bag Placement: Operator places an empty bag under the chute. The system is armed for filling.
3. Filling: Auger motor is activated. PLC continuously monitors load cell.
4. Target Weight Reached: Auger stops at 25 kg ± 1
5. Conveying: Conveyor motor starts, moving the filled bag toward the sealing station.
6. Sealing: Band sealer closes the bag at preset temperature and dwell time.
7. Discharge: Conveyor transports sealed bag to the end for manual collection.
8. Reset: system returns to idle and awaits the next cycle.

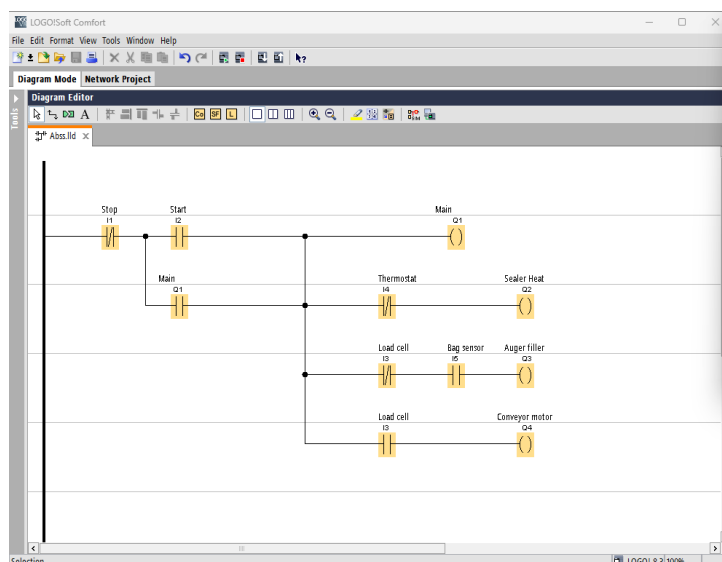


Figure 3.4: ladder diagram

Safety conditions, such as emergency stop signals, overload protection, and heater over-temperature, are handled in parallel routines that override the normal sequence if triggered.

3.4.6 Literature Review

Packaging automation has become a critical component in modern manufacturing industries. It plays a vital role in ensuring consistency, efficiency, and quality in product handling and delivery. Traditional manual packaging methods, though simple and inexpensive, are associated with inefficiency, labor intensity, and inconsistency in product quality. Literature indicates that automation in packaging provides solutions to these shortcomings by introducing systems that are precise, reliable, and cost-effective.

1. **Automated Bagging Systems** Automated bagging systems are increasingly used in agro-processing industries, particularly in the packaging of grain, flour, and animal feed. According to Adepoju et al. such systems improve productivity by reducing reliance on human labor while simultaneously increasing throughput. Bagging systems typically integrate weighing mechanisms, filling mechanisms, and sealing units, which work together under a centralized control system.
2. **Weighing Technologies** Accurate weighing is the cornerstone of automated packaging. Load cells are the most commonly used sensors due to their high accuracy and reliability. research by Zhang et al. **zhang2018** highlights that single-point load cells provide superior accuracy for bagging systems, reducing underfill and overfill errors. Integration with PLCs further enhances precision, allowing for automatic feedback control during the filling process.
3. **Bag Sealing Techniques** The most common method of closing polypropylene bags is manual stitching. However, Kumar and Singh **kumar2019** reported that manual stitching introduces variability in seal quality, dust leakage, and spillage, reducing the commercial value of products. Heat sealing, on the other hand, provides hermetic seals by fusing polymer layers under controlled temperature and

pressure. Patel et al. **patel2020** emphasize that continuous band sealers are particularly effective in maintaining hygiene and preventing contamination in food and feed packaging.

4. **PLC-Based Automation** Programmable Logic Controllers (PLCs) have become the backbone of industrial automation. They offer real-time processing, reliability, and flexibility for integrating sensors and actuators. In packaging plants, PLCs coordinate sequential operations such as filling, conveying, and sealing. As reported by Bandyopadhyay, PLC-based automation reduces human error, increases efficiency, and improves the safety of industrial processes.

5. **Summary** The reviewed literature demonstrates that:

- Manual bag closing is inefficient and inconsistent.
- Heat sealing provides a more durable and hygienic closure than stitching.
- Load cells integrated with PLCs improve weighing accuracy and reliability.
- Automation in packaging reduces labor requirements, increases throughput, and enhances product quality.

This project builds on these findings by designing an Automated Bagging and Sealing System (ABSS) tailored for Bejai Ethio Industrial & Engineering Solutions.

3.5 Results and Discussion

3.6 Results

The project involved the design of an Automated Bagging and Sealing System that integrates weighing, filling, conveying, and sealing into a single automated process. The main results are:

1. **Mechanical Design:** A 3 m conveyor was designed to transport filled bags at a speed of 0.067 m/s. A 0.75 kW auger motor was selected to fill 25 kg of animal feed into each bag within 20 seconds. A continuous band sealer was incorporated to provide uniform sealing at a dwell time of 1.2–1.5 s and temperature of 170 °C.
2. **Electrical Design:** A 50 kg stainless-steel single-point load cell was chosen to measure bag weight accurately. Motors were selected as follows: conveyor motor (0.37 kW), auger motor (0.75 kW), and sealer band motor (0.1 kW). A PLC was used as the central control unit to synchronize the filling, conveying, and sealing operations.
3. **Software Design:** A control logic was developed to start filling when a bag is placed, monitor weight through the load cell, stop filling at 25 kg, and activate the conveyor and sealer in sequence. Safety features such as emergency stop, overload protection, and over-temperature alarms were integrated.

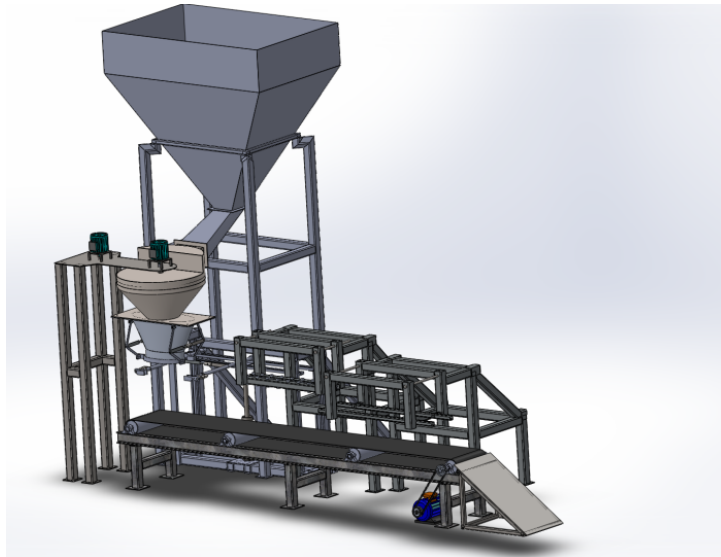


Figure 3.5: Final Assembly Design

3.7 Discussion

The proposed system addresses the key challenges identified in the company:

- **Improved Accuracy:** The use of a load cell ensures precise filling, eliminating inconsistencies caused by manual weighing.
- **Better Seal Quality:** Heat sealing produces dust-free, tamper-proof seals compared to manual stitching, improving hygiene and presentation.
- **Labor Reduction:** Manual bagging and sealing requires 3–4 workers, while the automated system requires only one operator.
- **Increased Productivity:** The proposed system achieves three bags per minute compared to the manual one-bag-per-minute rate, effectively tripling throughput.
- **Operational Reliability:** Automation reduces dependence on labor availability and minimizes downtime due to fatigue or absenteeism.

Although the system requires initial investment in machinery and training, its long-term benefits in reduced labor costs, increased throughput, and improved product quality outweigh the challenges.

3.8 Conclusion

The internship project focused on the design of an Automated Bagging and Sealing System (ABSS) for Bejai Ethio Industrial & Engineering Solutions. The study identified that the existing manual bagging and sealing system was labor-intensive, inconsistent, and inefficient.

The proposed system integrates:

- A load cell-based weighing system for precision.
- An auger filler for consistent material dispensing.

- A motorized conveyor for smooth bag transfer.
- A continuous band sealer for durable, hygienic packaging.

The results showed that automation significantly reduces labor requirements, improves accuracy, enhances seal quality, and increases productivity. By moving from manual stitching to automated sealing, the company can achieve dust-free, uniform, and market-ready packaging.

In conclusion, the project demonstrates how engineering principles can be applied to solve real-world industrial problems. Implementation of this system would provide Bejai with improved operational efficiency, reduced costs, and enhanced product quality. Future work should focus on fabricating a prototype, testing performance under real operating conditions, and refining design parameters for optimal operation.

This experience honed our ability to diagnose industrial problems and engineer practical, sustainable solutions.

4 Overall Internship Achievements and Experiences

During our internship at Bejai Ethio Industrial & Engineering Solutions, we were deeply involved in the design and development of an Automated Bagging and Sealing System (ABSS). This experience allowed us to apply theoretical knowledge to real-world engineering challenges, significantly enhancing both our technical and professional skills. Below is a summary of our key achievements and learning experiences.

4.1 Theoretical Knowledge Enhancement

Through the design and analysis of the ABSS, we strengthened our understanding of core engineering principles:

- **Mechanical Design:** We applied concepts of conveyor dynamics, torque calculation, power transmission, and material selection to design a robust bag-handling system.
- **Electrical Systems:** We deepened our knowledge of motor control, sensor integration (load cell, photoelectric sensors), and power distribution systems.
- **Control Systems:** We gained insight into PLC programming, automation sequencing, and feedback control mechanisms for precise weighing and sealing.
- **Thermodynamics:** Understanding heat transfer principles was essential for designing the band sealer to achieve consistent and reliable seals.

This project reinforced our academic learning and demonstrated how interdisciplinary knowledge is essential in solving complex industrial problems.

4.2 Practical Skill Development

The hands-on nature of the project allowed us to develop valuable practical skills:

- **CAD and Modeling:** We used SolidWorks for mechanical design and simulation, including conveyor layout, auger filler mechanism, and structural framing.

- **Circuit Design:** We designed electrical schematics for motor control, sensor interfacing, and safety circuits compatible with 220V AC supply.
- **Programming:** We developed ladder logic for PLC-based control systems to automate the filling, conveying, and sealing processes.
- **Prototyping and Testing:** Although full fabrication was beyond the scope, we performed virtual simulations and parameter validation for weight accuracy, sealing temperature, and conveyor speed.

These skills are directly transferable to future engineering roles and provided a solid foundation for professional practice.

4.3 Industrial Problem-Solving Capability

The ABSS project was initiated to address a clear operational bottleneck. Through systematic observation, analysis, and design, we:

- Identified root causes of inefficiencies in the manual bagging process.
- Proposed an integrated automated solution that reduces labor, improves accuracy, and increases throughput.
- Considered real-world constraints such as power availability, material properties, and operational safety.
- Developed a feasible, cost-effective design tailored to the company's production environment.

This experience honed our ability to diagnose industrial problems and engineer practical, sustainable solutions.

4.4 Teamwork and Collaboration Skills

Working in a group to design the ABSS required effective collaboration and communication:

- We divided tasks based on individual strengths—mechanical design, electrical systems, control logic, and documentation.
- Regular meetings ensured alignment on design parameters, progress updates, and problem-solving.
- We learned to incorporate feedback from both university and industry supervisors, improving the design iteratively.
- This collaborative approach mirrored real-world engineering projects, preparing us for professional teamwork.

4.5 Understanding of Work Ethics

The internship instilled a strong sense of professional responsibility and ethics:

- We adhered to safety standards in all design considerations, including emergency stops, overload protection, and thermal safeguards.
- We maintained integrity in data collection, analysis, and reporting, ensuring accuracy and transparency.
- We learned the importance of deadlines, accountability, and delivering quality work under constraints.

These values are essential for any engineering professional and were reinforced throughout the project.

4.6 Entrepreneurship and Industry Insight

The project exposed us to the business and operational aspects of industrial automation:

- We understood how automation can reduce operational costs, improve product quality, and enhance market competitiveness.
- We gained insight into the economic feasibility of engineering solutions, considering initial investment vs. long-term benefits.
- The experience encouraged an innovative mindset, looking for opportunities to improve processes through technology.

This broader perspective helps bridge the gap between technical design and business impact.

5 Conclusion

The internship at Bejai Ethio Industrial & Engineering Solutions provided us with invaluable exposure to real-world industrial electromechanical systems. It allowed us to bridge the gap between theory and practice by applying knowledge gained in courses such as Strength of Materials, Machine Elements, Industrial Automation, Control Systems, and Thermodynamics. Through direct involvement in system observation, design, and problem-solving, we gained hands-on skills in PLC programming, mechanical design, and integration of electrical and mechanical systems.

Beyond technical growth, the internship strengthened our professionalism in teamwork, communication, ethics, and industry awareness. We were able to observe the complete process of project development, beginning with market research and feasibility studies, followed by design, workshop fabrication, assembly, and performance testing. This comprehensive experience enriched both our design knowledge and our practical workshop skills, as we collaborated closely with engineers and technicians.

Overall, the internship highlighted the multidisciplinary nature of industrial engineering work, where effective collaboration across different fields is essential. It prepared us not only to solve technical problems but also to contribute meaningfully to real-world engineering projects with professionalism, creativity, and adaptability.

5.1 Recommendations

To Bejai Ethio Industrial & Engineering Solutions

We recommend that the company provide more structured mentorship for interns. This could include assigning specific mentors to guide interns on functional skills and tasks relevant to the internship objectives. Additionally, offering multiple projects within the interns' capabilities would enhance their learning experience and practical exposure.

To Addis Ababa Science and Technology University (AASTU)

We suggest that assigned faculty members engage more actively with interns, conducting regular check-ins every three weeks to assess progress and discuss lessons learned. Furthermore, incorporating more industrial communication protocols into the curriculum would better prepare students for real-world engineering challenges.

To Future Interns

We advise future interns to conduct thorough background research on the company before starting their internship. Actively engaging in the internship experience and striving to learn as much as possible will greatly benefit their future careers. Additionally, maintaining punctuality and showing respect toward teammates are essential for fostering a professional and collaborative work environment.